



تعميم

المحترمون

السادة/

السلام عليكم ورحمة الله وبركاته،

الموضوع: تحديث السيناريوهات المعيارية لصدمات أسعار الفائدة في الدفاتر البنكية

استناداً إلى الصلاحيات المخولة للبنك المركزي السعودي بموجب نظامه الصادر بالمرسوم الملكي رقم (م/٣٦) وتاريخ ١٤٤٢/٤/١١هـ، ونظام مراقبة البنوك الصادر بالمرسوم الملكي رقم (م/٥) وتاريخ ١٣٨٦/٢/٢٢هـ وإحاقاً إلى التعميم رقم (٣٨١٠٠٠٤٠٢٤٣) وتاريخ ١٤٣٨/٤/١٢هـ والمتعلق بمبادئ إدارة مخاطر أسعار الفائدة في الدفاتر البنكية (IRRBB).

نفيدكم باعتماد التحديث للسيناريوهات المعيارية لصدمات أسعار الفائدة ليكون وفق الصيغة المرفقة، وذلك بما يتماشى مع التحديثات ذات الصلة الصادرة عن لجنة بازل للرقابة المصرفية.

للاحاطة والعمل بموجبه ابتداءً من ١ يناير ٢٠٢٦ م.

الزييد

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وكيل المحافظ للرقابة

نطاق التوزيع:

- البنوك والمصارف المحلية العاملة في المملكة.

Recalibration of shocks in the interest rate risk in the banking book

The Standardised Interest Rate Shock Scenarios

1. Banks should apply six prescribed interest rate shock scenarios to capture parallel and nonparallel gap risks for EVE and two prescribed interest rate shock scenarios for NII. These scenarios are applied to IRRBB exposures in each currency for which the bank has material positions. The six shock scenarios reflect currency-specific absolute shocks as specified in Table 1 below. Under this approach, IRRBB is measured by means of the following six scenarios:

- 1) Parallel shock up;
- 2) Parallel shock down;
- 3) Steepener shock (short rates down and long rates up);
- 4) Flatteners shock (short rates up and long rates down);
- 5) Short rates shock up; and
- 6) Short rates shock down.

$\bar{S}_{shocktype,c}$

Specified Size of Interest Rate Shocks			Table 1
Currency	Shock Scenarios		
	Parallel	Short	Long
ARS	400	500	300
AUD	350	425	300
BRL	400	500	300
CAD	200	275	175
CHF	175	250	200
CNY	225	300	150
EUR	225	350	200
GBP	275	425	250
HKD	225	375	200
IDR	400	500	300
INR	325	475	225
JPY	100	100	100
KRW	225	350	225
MXN	400	500	200
RUB	400	500	300
SAR	275	375	250
SEK	275	425	200
SGD	175	250	225
TRY	400	500	300
USD	200	300	225
ZAR	325	500	300

2. Given Table 1 above, the instantaneous shocks to the risk-free rate for parallel, short and long, for each currency, the following parameterisations of the six interest rate shock scenarios should be applied:

- (1) Parallel shock for currency c : a constant parallel shock up or down across all time buckets.

$$\Delta S_{parallel,c}(t_k) = \pm \bar{S}_{parallel,c}$$

- (2) Short rate shock for currency c : shock up or down that is greatest at the shortest tenor midpoint. That shock, through the shaping scalar

$$\alpha_{short}(t_k) = e^{\frac{-t_k}{x}}$$

where $x = 4$, diminishes towards zero at the tenor of the longest point in the term structure.¹

$$\Delta S_{short,c}(t_k) = \pm \bar{S}_{short,c} \cdot \alpha_{short}(t_k) = \pm \bar{S}_{short,c} \cdot e^{\frac{-t_k}{x}}$$

- (3) Long rate shock for currency c (note: this is used only in the rotational shocks): Here the shock is greatest at the longest tenor midpoint and is related to the short scaling factor as:

$$\alpha_{long}(t_k) = 1 - \alpha_{short}(t_k)$$

$$\Delta S_{long,c}(t_k) = \pm \bar{S}_{long,c} \cdot \alpha_{long}(t_k) = \pm \bar{S}_{long,c} \cdot \left(1 - e^{\frac{-t_k}{x}}\right)$$

- (4) Rotation shocks for currency c : involving rotations to the term structure (i.e. steepeners and flatteners) of the interest rates whereby both the long and short rates are shocked and the shift in interest rates at each tenor midpoint is obtained by applying the following formulas to those shocks:

$$\Delta S_{steepener,c}(t_k) = -0.65 \cdot |\Delta S_{short,c}(t_k)| + 0.9 \cdot |\Delta S_{long,c}(t_k)|$$

$$\Delta S_{flattener,c}(t_k) = +0.8 \cdot |\Delta S_{short,c}(t_k)| - 0.6 \cdot |\Delta S_{long,c}(t_k)|$$

3. The floors for the post-shock interest rates under the six interest rate shock scenarios are set at zero. However, if circumstances change in future, SAMA may revise it accordingly.

⁽¹⁾ The value of x in the denominator of the function $\frac{-t_k}{x}$ controls the rate of decay of the shock. This should be set to the value of 4 for most currencies and the related shocks unless otherwise determined by SAMA. t_k is the midpoint (in time) of the k^{th} bucket and t_k is the midpoint (in time) of the last bucket K . There are 19 buckets in the standardised framework, but the analysis may be generalised to any number of buckets.

4. The following examples illustrate the scenarios in (2) and (4) above.

- (1) Short rate shock: Assume that the bank uses the standardised framework with $K = 19$ time bands and with $t_k = 25$ years (the midpoint (in time) of the longest tenor bucket K), and where t_k is the midpoint (in time) for bucket k . In the standardised framework, if $k = 10$ with $t_k = 3.5$ years, the scalar adjustment for the short shock would be

$\alpha_{short}(t_k) = e^{\frac{-3.5}{4}} = 0.417$. Banks would multiply this by the value of the short rate shock to obtain the amount to be added to or subtracted from the yield curve at that tenor point. If the short rate shock was +100 basis points (bp), the increase in the yield curve at $t_k = 3.5$ years would be 41.7 bp.

- (2) Steepener: Assume the same point on the yield curve as above, $t_k = 3.5$ years. If the absolute value of the short rate shock was 100 bp and the absolute value of the long rate shock was 100 bp (as for the Japanese yen), the change in the yield curve at $t_k = 3.5$ years would be the sum of the effect of the short rate shock plus the effect of the long rate shock in bp: $-0.65 \times 100 \text{ bp} \times 0.417 + 0.9 \times 100 \text{ bp} \times (1 - 0.417) = +25.4 \text{ bp}$.

- (3) Flattener: The corresponding change in the yield curve for the shocks in the example above at $t_k = 3.5$ years would be: $+0.8 \times 100 \text{ bp} \times 0.417 - 0.6 \times 100 \text{ bp} \times (1 - 0.417) = -1.6 \text{ bp}$.